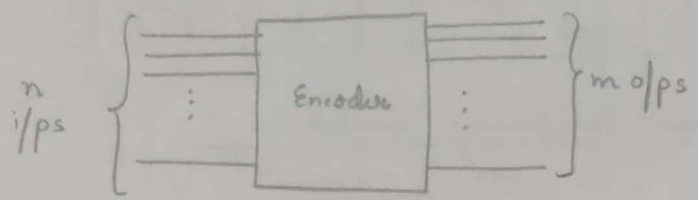
Encoders

An encoder is a digital circuit that performs the inverse operation of a decoder. Hence, the opposite of the decoding process is called encoding.

An encoder is a combinational logic circuit that converts an active i/p signal into a coded o/p signal.

It has n i/p lines, only one of which is active at any time & m o/p lines. It encodes one of the active i/ps to a coded binary o/p with m bits.

In an encoder, no. of o/ps is less than the no. of i/ps.



$(m<n)$

Block diagram of an encoder

Octal-to-Binary Encoder -

Binary-to-octal decoder (3-to-8 decoder) accepts a 3-bit i/p code & activates one of 8 o/p lines corresponding to that code.

An octal-to-binary encoder performs the opposite function, it accepts 8 i/ps & produces a 3-bit o/p code corresponding to the activated i/p.

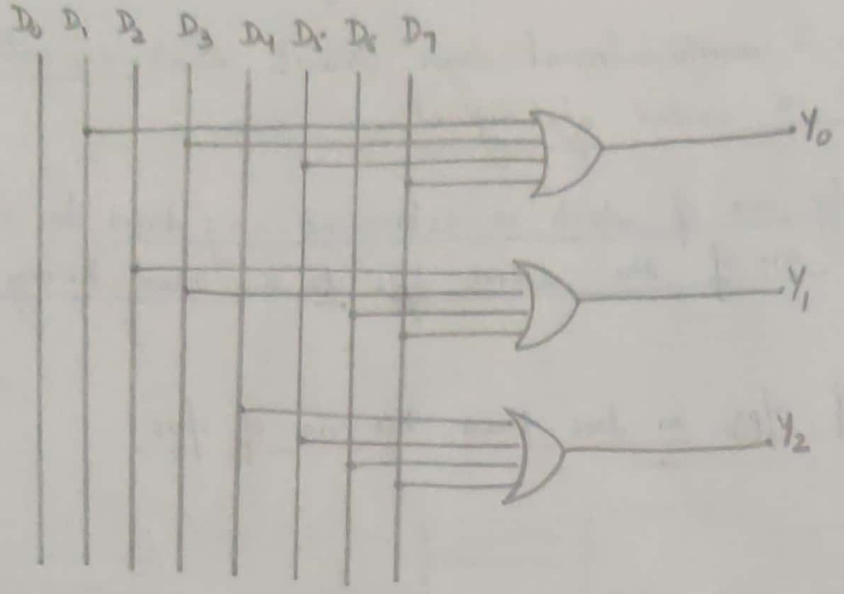
The truth table is shown below:-

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	Y_2	Y_1	Y_0
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

We have, $Y_0 = D_1 + D_3 + D_5 + D_7$

Similarly, $Y_1 = D_2 + D_3 + D_6 + D_7$

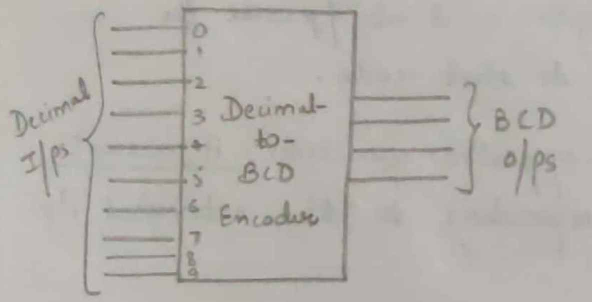
$Y_2 = D_4 + D_5 + D_6 + D_7$



Octal-to-binary encoder

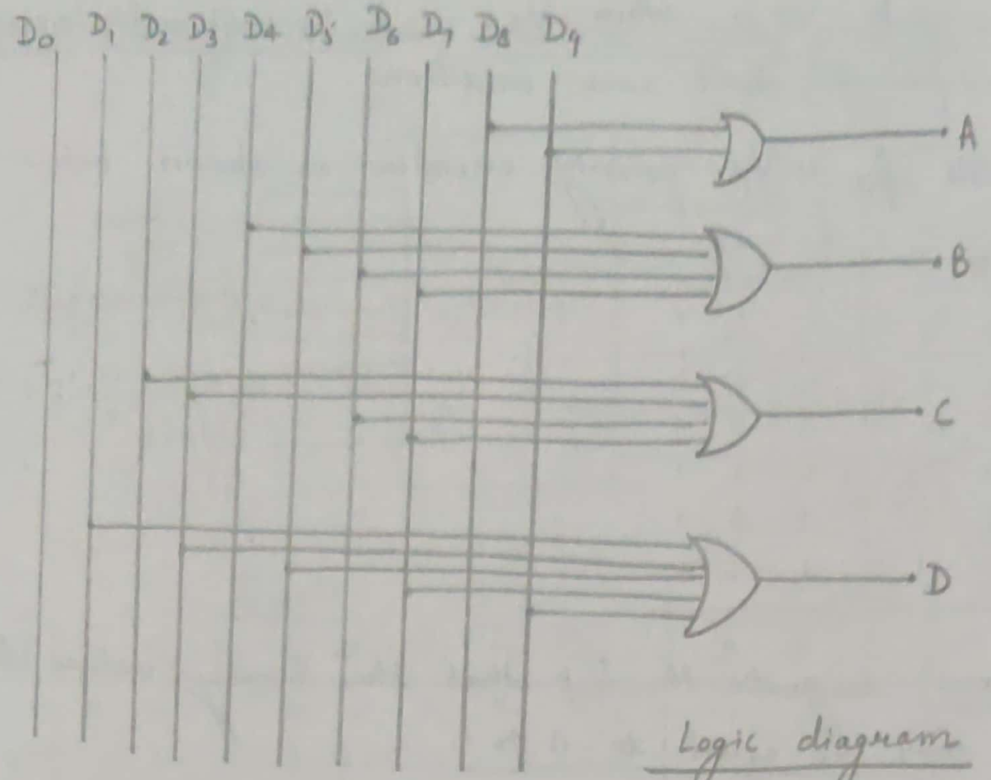
Decimal-to-BCD Encoder -

A decimal to BCD encoder has 10 i/ps corresponding to 10 decimal digits (0 to 9) & 4 o/ps (A, B, C, D) representing BCD value of i/p decimal digit.



Decimal I/ps										BCD o/ps			
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	A	B	C	D
1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	0	0	0	1	0	0
0	0	0	0	0	1	0	0	0	0	0	1	0	1
0	0	0	0	0	0	1	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	1	1	1
0	0	0	0	0	0	0	0	1	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1	1	0	0	1

$$\begin{cases} A = D_8 + D_9 \\ B = D_4 + D_5 + D_6 + D_7 \\ C = D_2 + D_3 + D_6 + D_7 \\ D = D_1 + D_3 + D_5 + D_7 + D_9 \end{cases}$$



Logic diagram

Priority Encoder

Encoders discussed so far will operate correctly, provided that only one i/p is HIGH at any given time.

But in some practical systems, 2 or more decimal i/ps may inadvertently become HIGH at the same time.

A priority encoder is a logic circuit that responds to just one i/p in accordance with some priority system, among all those that may be simultaneously HIGH.

The most common priority system is based on the relative magnitudes of the i/ps; whichever decimal point is largest that i/p is encoded.

4- i/p priority encoder-

In addition to the o/ps Y_2 & Y_1, the circuit has a 3rd o/p designated by V. This is a valid bit indicator that is set to 1 when one or more i/ps are equal to 1. If all i/p = 0, there is no valid i/p & V = 0. When V = 0, Y_2 & Y_1 are not inspected & are specified as don't care conditions.

Truth table for 4-i/p priority encoder is shown below:-

Inputs				Outputs		
D_0	D_1	D_2	D_3	Y_2	Y_1	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

$Y_2 = \Sigma m(1,2,3,5,6,7,9,10,11,13,14,15)$

& $Y_1 = \Sigma m(1,3,4,5,7,9,11,12,13,15)$

* X (Don't cares) designate the fact that the binary values they represent may be equal to 0 or 1.
* Also, D_3 has highest priority; regardless of the values of other i/ps, when $D_3 = 1$, $Y_2 Y_1 = 11$.
* D_2 has next priority level. O/p is 10 if $D_2 = 1$, provided that $D_3 = 0$ regardless of values of other two lower priority i/ps.
* D_1 is generated only when higher priority i/ps are 0 & so on.

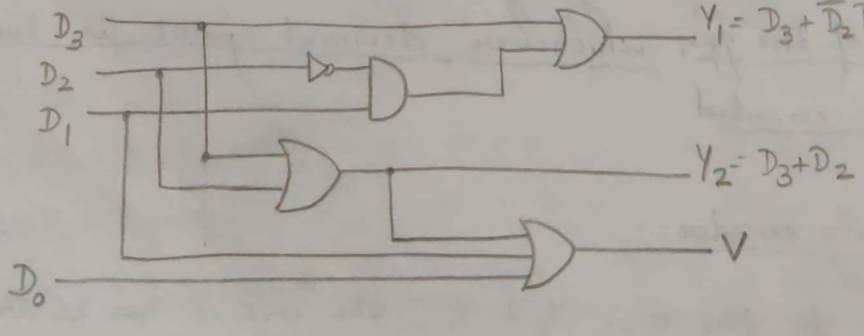
From the truth table,

$$Y_2 = D_3 + \overline{D_3} D_2 = D_3 + D_2$$

$$Y_1 = D_3 + \overline{D_3}\,\overline{D_2}\,D_1 = D_3 + \overline{D_2} D_1$$

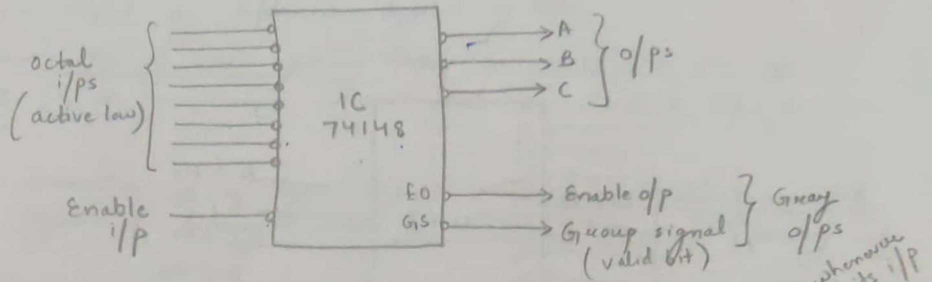
(Solve by k-maps)

$$V = D_3 + D_2 + D_1 + D_0$$



Logic diagram

8-to-3 Priority Encoder (Octal-to-binary priority encoder)

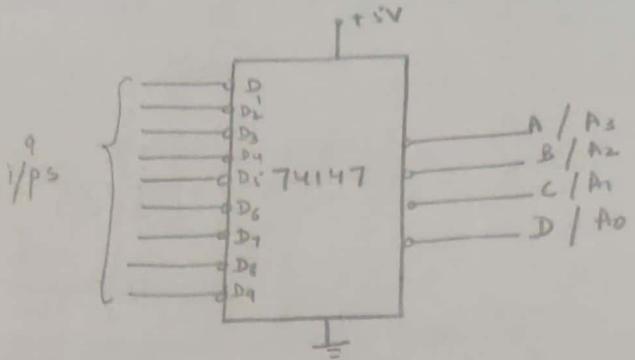


Inputs									Outputs				
EI	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	A	B	C	GS	EO
1	X	X	X	X	X	X	X	X	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	0
0	X	X	X	X	X	X	X	0	0	0	0	0	1
0	X	X	X	X	X	X	0	1	0	0	1	0	1
0	X	X	X	X	X	0	1	1	0	1	0	0	1
0	X	X	X	X	0	1	1	1	0	1	1	0	1
0	X	X	X	0	1	1	1	1	1	0	0	0	1
0	X	X	0	1	1	1	1	1	1	0	1	0	1
0	X	0	1	1	1	1	1	1	1	1	0	0	1
0	0	1	1	1	1	1	1	1	1	1	1	0	1

Decimal-to-BCD Priority Encoder-

Decimal I/ps										BCD o/ps			
~~D_0~~	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	A_3	A_2	A_1	A_0
1	1	1	1	1	1	1	1	1	1	1	1	1	1
X	X	X	X	X	X	X	X	X	0	0	1	1	0
X	X	X	X	X	X	X	X	0	1	0	1	1	1
X	X	X	X	X	X	X	0	1	1	1	0	0	0
X	X	X	X	X	X	0	1	1	1	1	0	0	1
X	X	X	X	X	0	1	1	1	1	1	0	1	0
X	X	X	X	0	1	1	1	1	1	1	0	1	1
X	X	X	0	1	1	1	1	1	1	1	1	0	0
X	X	0	1	1	1	1	1	1	1	1	1	0	1
X	0	1	1	1	1	1	1	1	1	1	1	1	0
~~[illegible]~~	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~

We observe that there are only 9 i/ps ($D_1 - D_9$). When all $D_1 - D_9$ are HIGH, o/ps are 1111 $\Rightarrow$ which is the BCD code for 0.



Block dig. of Decimal-to-BCD priority encoder